UW-Madison GI Tract Image Segmentation

Track Healthy Organs in Medical Scans to Improve Cancer Treatment

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Spring 2026

Agenda

1. Introduction
2. Challenge Objective
3. Dataset Overview
4. Data Challenges
5. Next Steps
6. Conclusion

Problem Overview

The Challenge

This challenge aims to **automate the segmentation of the stomach and bowels** in MRI scans to enhance the precision of cancer radiotherapy.

The Provider

Hosted by the **UW-Madison Carbone Cancer Center**, which provided anonymized MRI scans from actual cancer patients.

The Importance

Enables **real-time adaptive radiotherapy**, allowing clinicians to deliver higher radiation doses to tumors while sparing healthy moving organs, ultimately reducing patient treatment time and toxic side effects.

Objective and Clinical Impact

Technical Goal

- **Task:** Pixel-wise semantic segmentation of GI tract organs.
- **Output:** Binary masks converted to **Run-Length Encoding (RLE)** for efficiency.
- **Success Metrics:** Dice Coefficient (overlap) & Hausdorff Distance (boundary precision).

The "MR-Linac" Revolution

Automates manual contouring, turning up to **60-minute processes** into **near-instant** execution.

Clinical Benefits:

- On-the-fly adjustments.
- Higher radiation precision.

Dataset Anatomy: Scale & Structure

Attribute	**Specification**
Total Images	38,496 MRI Slices
Case Studies	85 Patients (Patient Data Spans Multiple Days)
File Format	16-bit Grayscale PNG
Variable Resolution	e.g., 266×320 to 360×310 px
Total Size	2.47 GB

Table: Also Contains a csv of all IDs and Masks of training objects

Class Distribution & Sparsity

Target Classes:

1. **Stomach**
2. **Small Bowel**
3. **Large Bowel**

Key Observation: Classes are **non-exclusive**. A single slice may contain one, all, or *none* of these organs.

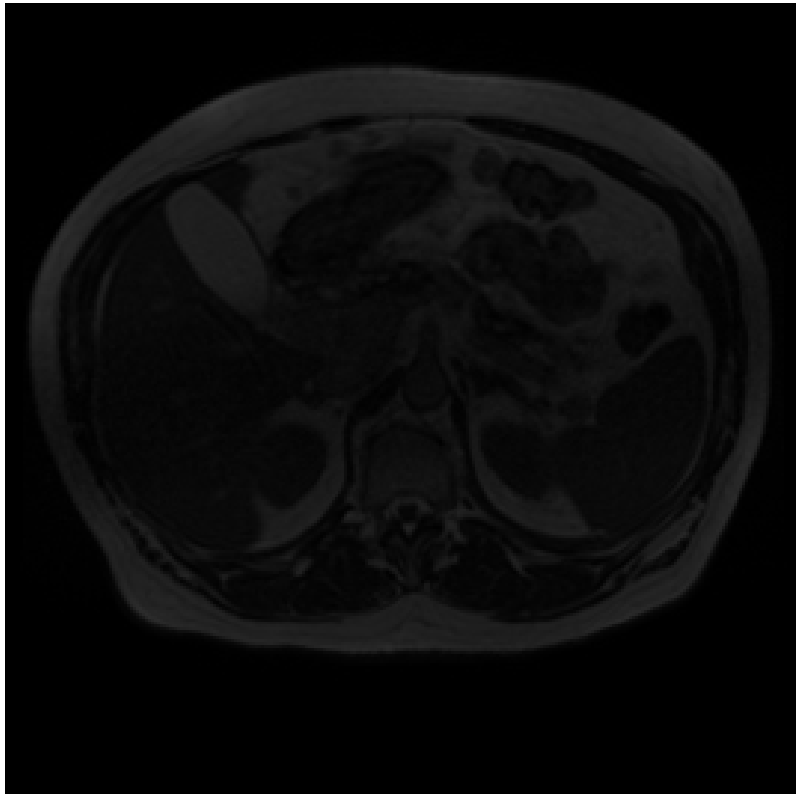
The Background Problem

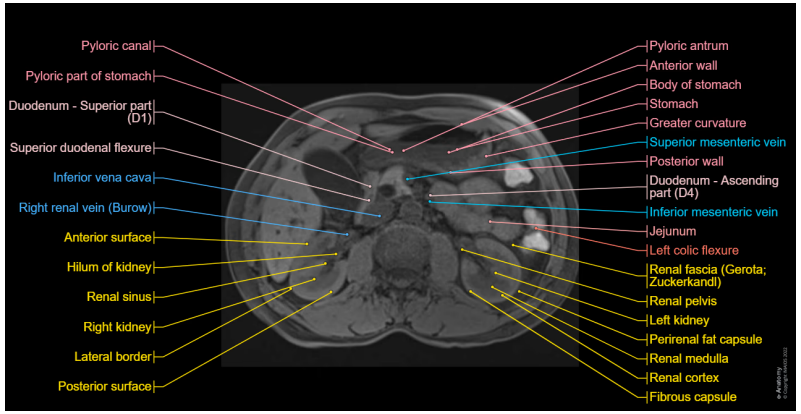
A significant portion of the dataset consists of **"Empty Slices"**, images with no labeled organs. The model must be robust against false positives in these regions.

Preprocessing Requirements

- **Normalization:** Converting 16-bit intensity to 8-bit or float32.
- **Resizing:** Standardizing variable resolutions to 256×256 or 320×320.

Example Data: MRI Scan Analysis





Key Observations:

- **Soft Tissue Contrast:** Subtle gray-level differences distinguish organs.
- **ROI:** The stomach and intestines appear as fluid or air-filled cavities.
- **Noise:** Visual artifacts require careful normalization.

Pixel Goal

Each pixel must be classified as: *Stomach*, *Large Bowel*, *Small Bowel*, or *Background*.

Data Challenges & Ethical Considerations

Technical Challenges

- **High Sparsity:** Many slices contain zero target organs (Background class dominance).
- **Boundary Ambiguity:** Low contrast between the stomach and small bowel walls.
- **Resolution Drift:** Variable image sizes require padding or interpolation.
- **Sensor Noise:** MRI artifacts and "ghosting" from patient movement.

Ethics & Privacy: Data consists of **anonymized** MIR patient data.

Next Steps: Proposed Approach

Phase 1: Preprocessing & Data Prep

- **Normalization:** Rescale 16-bit MRI pixel values for model compatibility.
- **Image Resizing:** Standardize all scans to a uniform input size.
- **Mask Decoding:** Convert RLE strings from the CSV into binary masks.

Phase 2: Model Selection (U-Net)

- **Precise Localization:** Uses an **Encoder-Decoder** structure to classify pixels at original resolution.
- **Skip Connections:** Direct paths that preserve **fine-grained spatial details**, preventing the "blurring" of organ boundaries.

Phase 3: Evaluation

- Compare predictions against given **Success Metrics**.

Questions & Discussion